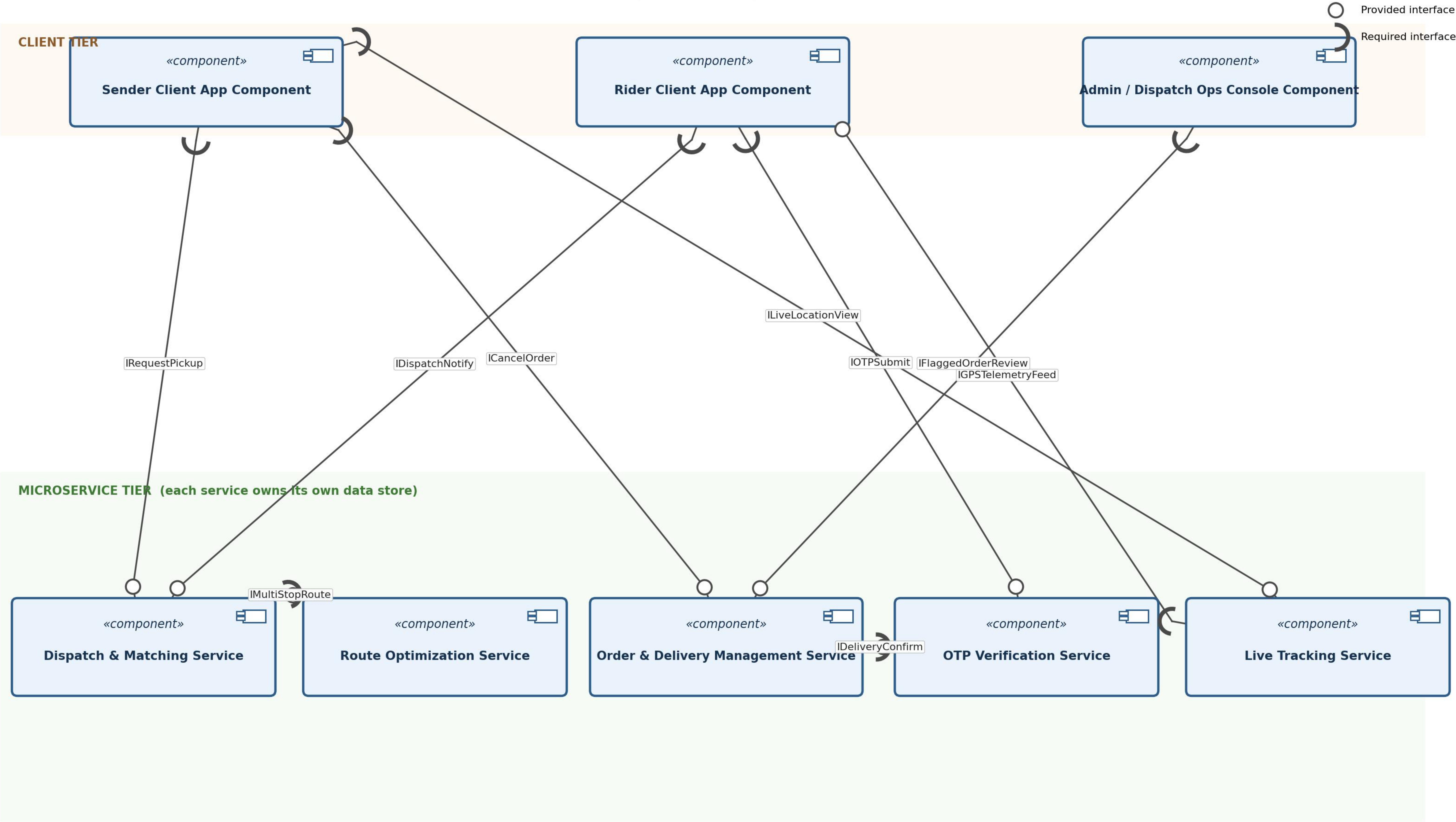


Hyperlocal Courier Dispatch & Tracking Engine — UML Component Diagram

Problem Statement #23 | Architectural Style: Microservices Architecture



Lab 3: Architectural Justification

Hyperlocal Courier Dispatch & Tracking Engine — Problem Statement #23

Architecture Selection: We chose Microservices Architecture for the Hyperlocal Courier Dispatch & Tracking Engine.

Architectural Choice

The system is decomposed into independently deployable services — Dispatch & Matching, Route Optimization, Order & Delivery Management, OTP Verification, and Live Tracking — each owning its own data store and exposed to the Sender Client, Rider Client, and Admin/Dispatch Ops Console through well-defined provided/required interfaces (REST and WebSocket).

Reasons for Selection

- **Wildly different scaling and latency profiles per requirement:** NFR-001 demands sub-2-second GPS telemetry delivery to every active sender, which means the Live Tracking service must handle high-frequency, low-latency streaming independently of everything else. Route Optimization, by contrast, is invoked only when a rider has ≥ 2 stops and can be compute-heavy per call (FR-003). A single layered monolith would force these opposite workloads to scale together; microservices let Live Tracking be scaled out for peak concurrent riders while Route Optimization scales separately on demand.
- **Independent evolution and fault isolation across FR-001–FR-005:** Dispatch & Matching (FR-001, NFR-002's 30-second reassignment timeout) and OTP Verification (FR-002) are logically and operationally distinct: if OTP Verification is slow or briefly down, in-flight matching and live tracking continue unaffected, and Order & Delivery Management simply holds the order at “In Transit” per the UC-05 alternate flow. This isolation matches the platform's core reliability requirement (NFR-002) far better than a single shared layer where one slow module can stall the whole request path.

Security Advantage

OTP validation logic and the 4-digit codes it compares against (FR-002) live entirely inside the OTP Verification service, reachable only through its IOTPSubmit interface. Neither the Rider Client nor the Order & Delivery Management service ever holds or computes the OTP directly — Order & Delivery Management only receives a signed IDeliveryConfirm result. This narrows the blast radius of a compromised client or service to a single, auditable boundary instead of exposing OTP comparison logic across the whole codebase.

Performance Benefit

Because Live Tracking is its own service, it can be deployed with a persistent WebSocket connection pool and scaled horizontally purely to meet NFR-001's under-2-second latency target, without competing for resources with the heavier, less frequent Route Optimization computations (FR-003) or the Dispatch & Matching service's nearest-rider queries (FR-001). Each service can be tuned and scaled to its own load pattern instead of a one-size-fits-all deployment.